

Day 09 – Linux User & Group Management Challenge

Task 1: Create Users

Create users with home directories

```
sudo useradd -m tokyo
sudo useradd -m berlin
sudo useradd -m professor
```

Set passwords

```
sudo passwd tokyo
sudo passwd berlin
sudo passwd professor
```

Verify users

```
cat /etc/passwd | grep -E "tokyo|berlin|professor"
ls -l /home
```

Task 2: Create Groups

```
sudo groupadd developers
sudo groupadd admins
```

Verify

```
cat /etc/group | grep -E "developers|admins"
```

Task 3: Assign Users to Groups

```
sudo usermod -aG developers tokyo
sudo usermod -aG developers,admins berlin
sudo usermod -aG admins professor
```

Verify group membership

```
groups tokyo
groups berlin
groups professor
```

Task 4: Shared Directory (/opt/dev-project)

Create directory

```
sudo mkdir /opt/dev-project
```

Set group owner

```
sudo chgrp developers /opt/dev-project
```

Set permissions (775)

```
sudo chmod 775 /opt/dev-project
```

Verify

```
ls -ld /opt/dev-project
```

Test as users

```
sudo -u tokyo touch /opt/dev-project/tokyo-file.txt  
sudo -u berlin touch /opt/dev-project/berlin-file.txt  
ls -l /opt/dev-project
```

Task 5: Team Workspace

Create user

```
sudo useradd -m nairobi  
sudo passwd nairobi
```

Create group

```
sudo groupadd project-team
```

Add users to group

```
sudo usermod -aG project-team nairobi  
sudo usermod -aG project-team tokyo
```

Create workspace

```
sudo mkdir /opt/team-workspace  
sudo chgrp project-team /opt/team-workspace  
sudo chmod 775 /opt/team-workspace
```

Verify

```
ls -ld /opt/team-workspace
```

Test as nairobi

```
sudo -u nairobi touch /opt/team-workspace/nairobi-file.txt  
ls -l /opt/team-workspace
```
